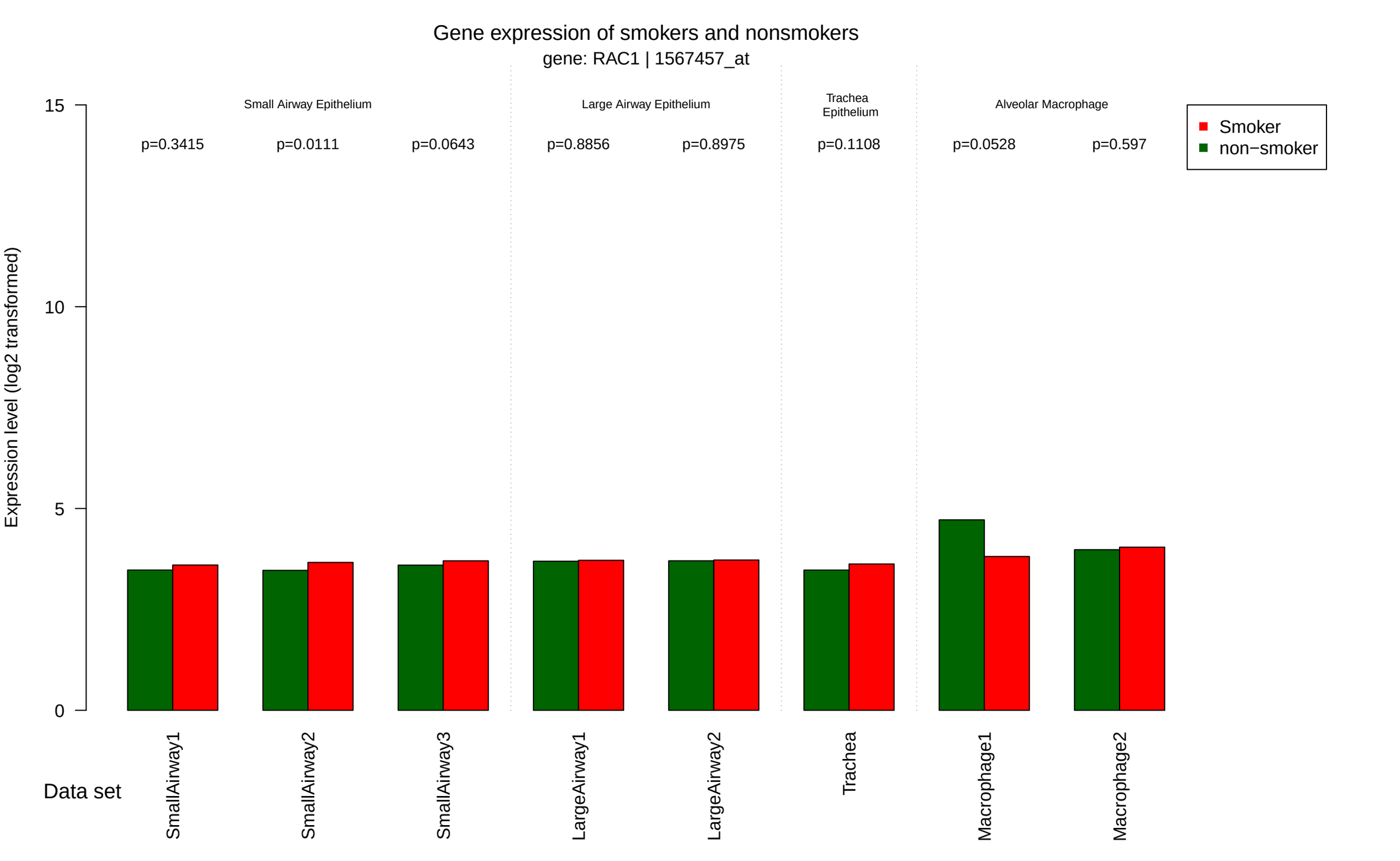
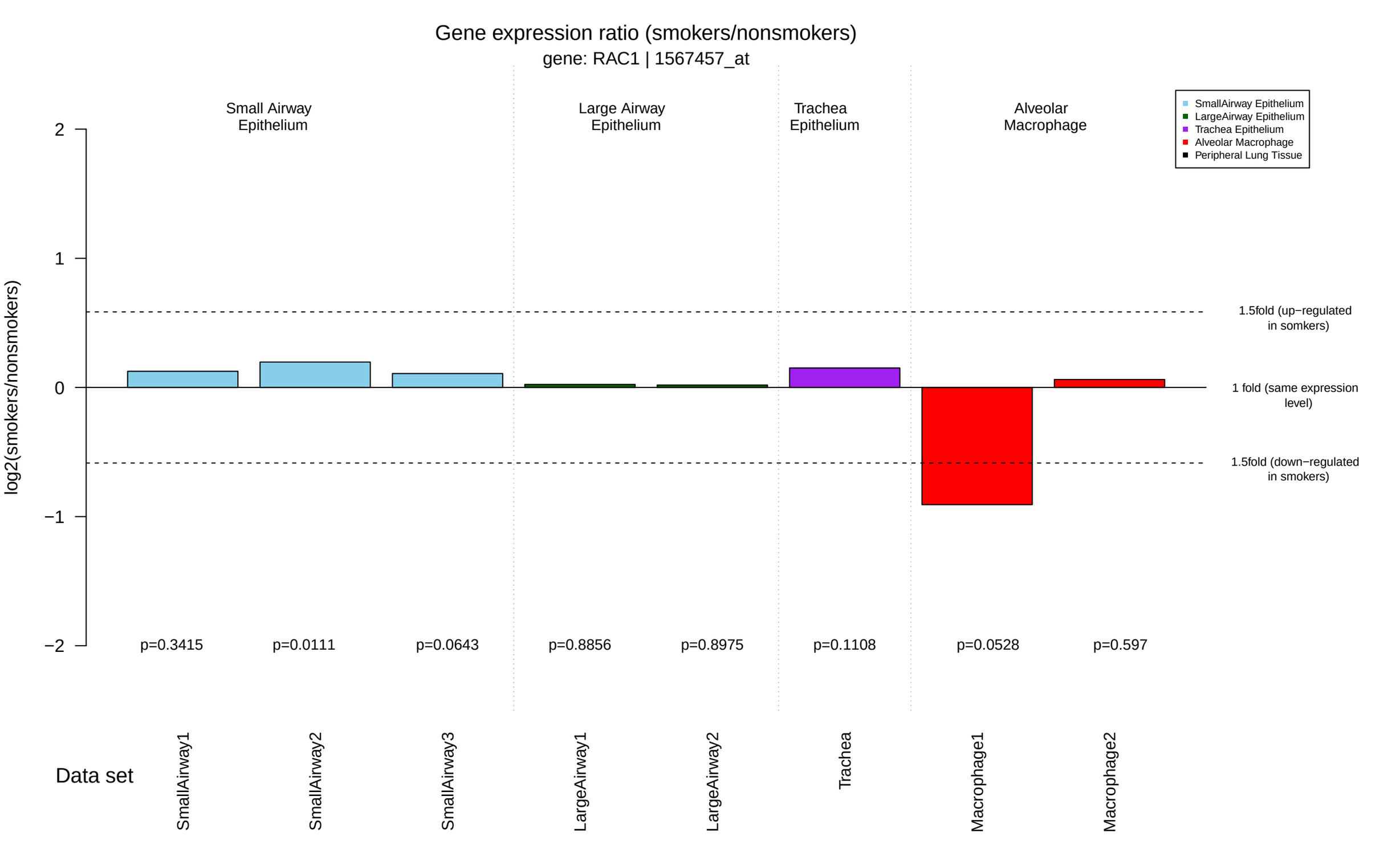
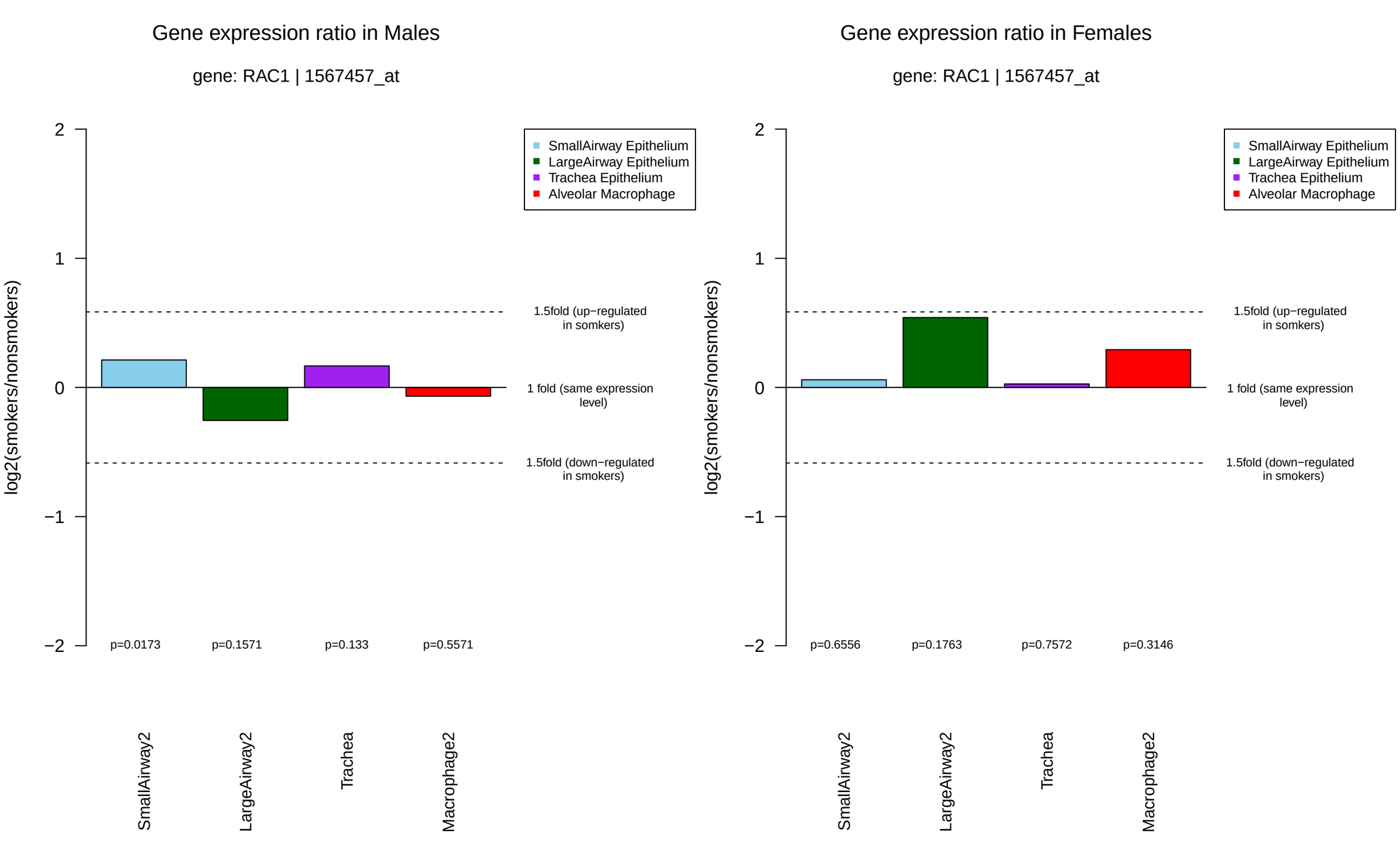
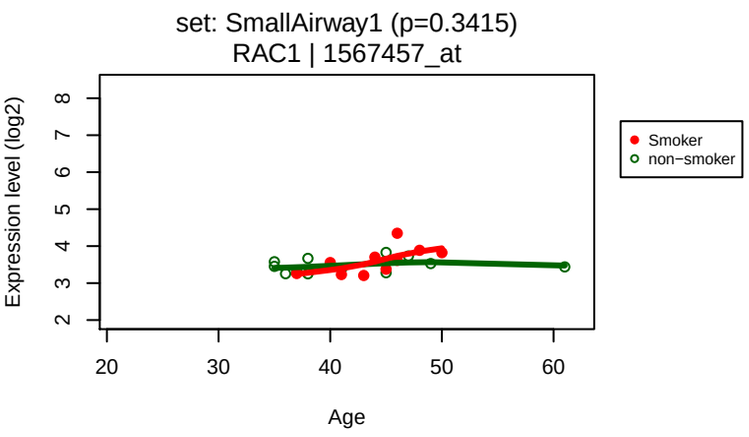
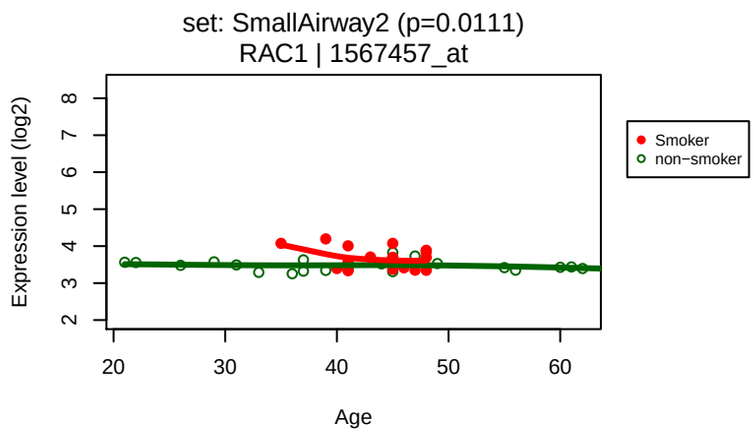
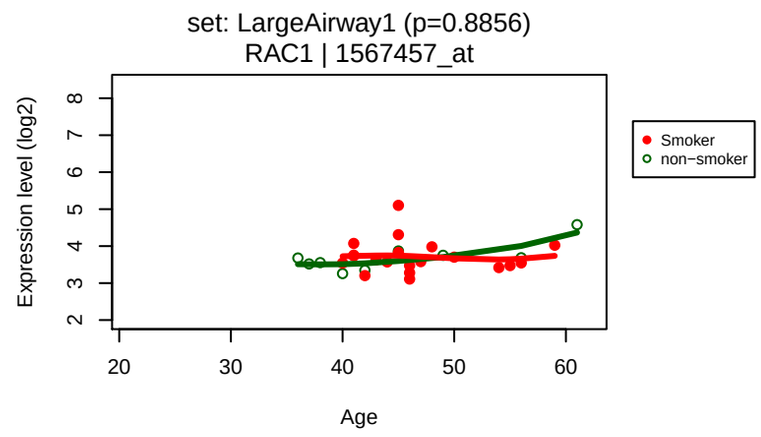

Gene expression of smokers and nonsmokers
gene: RAC1 | 1567457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3415
p=0.0111
p=0.0643
p=0.8856
p=0.8975
p=0.1108
p=0.0528
p=0.597
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

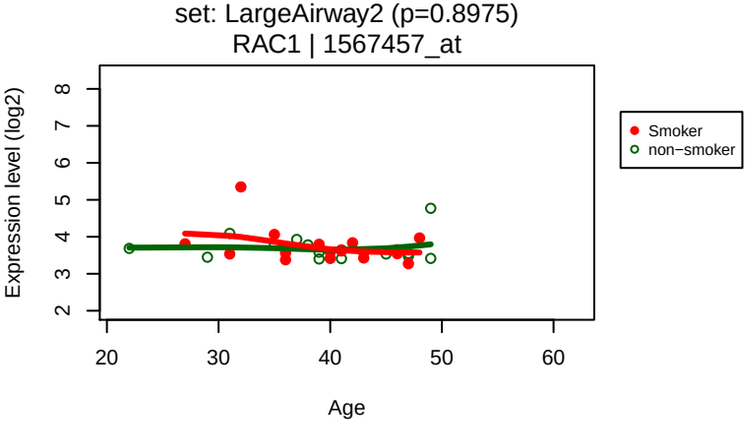

Gene expression ratio (smokers/nonsmokers)
gene: RAC1 | 1567457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3415
p=0.0111
p=0.0643
p=0.8856
p=0.8975
p=0.1108
p=0.0528
p=0.597
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

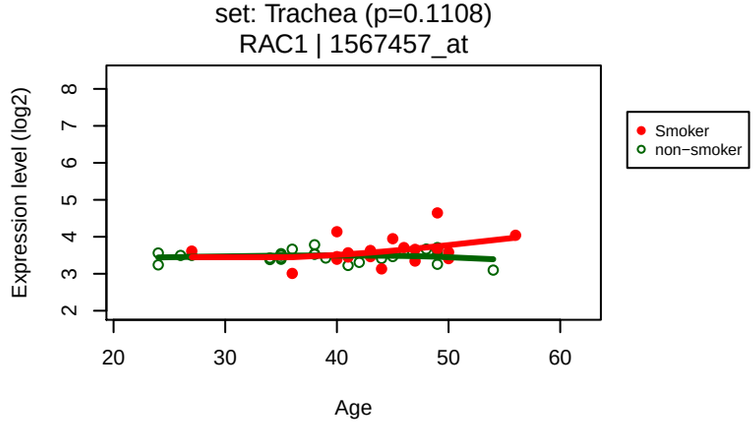

Gene expression ratio in Males
gene: RAC1 | 1567457_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0173
p=0.1571
p=0.133
p=0.5571
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC1 | 1567457_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6556
p=0.1763
p=0.7572
p=0.3146
SmallAirway2
LargeAirway2
Trachea
Macrophage2

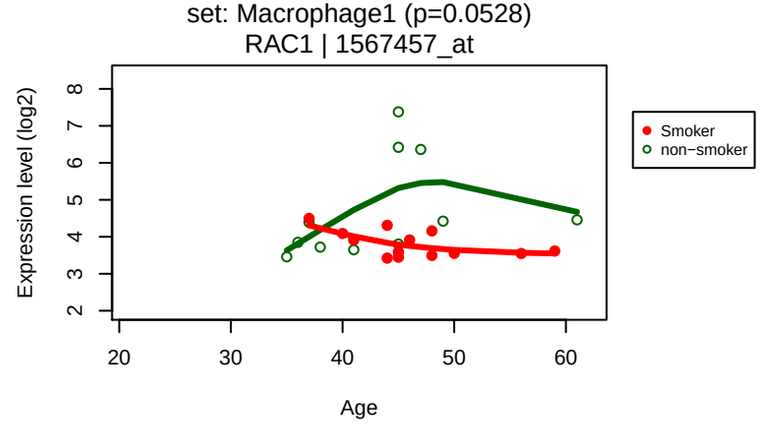

set: SmallAirway1 (p=0.3415)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker

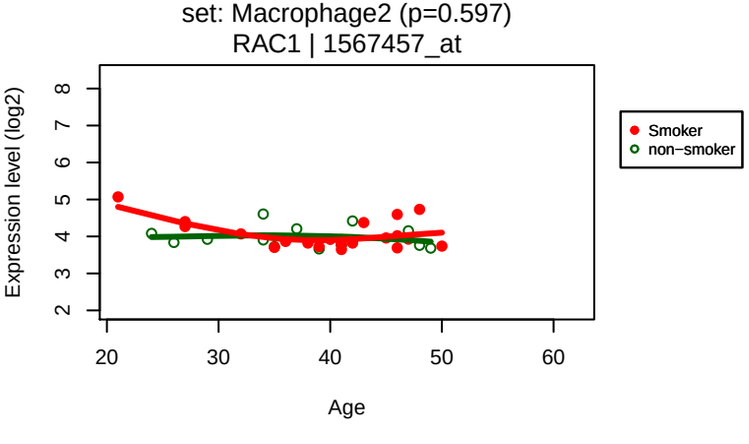

set: SmallAirway2 (p=0.0111)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker

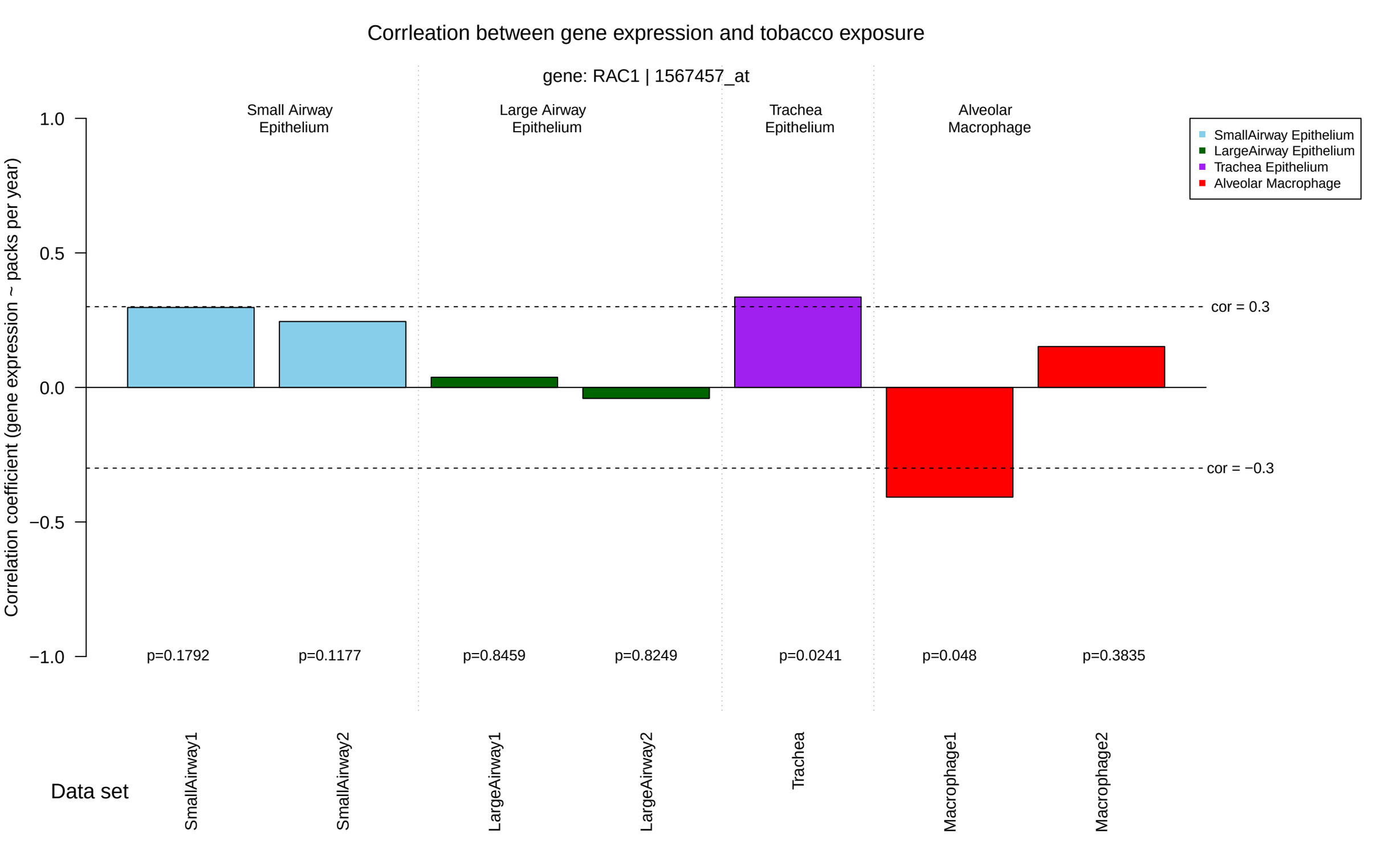

set: LargeAirway1 (p=0.8856)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8975)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1108)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0528)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.597)
RAC1 | 1567457_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC1 | 1567457_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1792
p=0.1177
p=0.8459
p=0.8249
p=0.0241
p=0.048
p=0.3835
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2